

Modern Cloud Computing and Distributed Systems

Core Service Models and Deployment Paradigms

Cloud computing delivers on-demand computing services including servers, storage, databases, networking, and software over the internet with pay-as-you-go pricing. The foundational service taxonomy spans:

- Infrastructure as a Service (IaaS): Virtualized compute and raw block storage (e.g., AWS EC2, GCP Compute Engine).
- Platform as a Service (PaaS): Managed execution environments that abstract operating system management (e.g., Heroku, AWS Elastic Beanstalk).

Microservices and Container Orchestration
Microservices are small, independently deployable applications delivered via web interfaces (e.g., Google Workspace, Salesforce).

Monolithic architectures often suffer from tight coupling, making independent scaling and continuous deployment difficult. Modern engineering teams decompose complex systems into decoupled microservices communicating through RESTful APIs, gRPC, or asynchronous message brokers such as Apache Kafka.

Containerization, standardizing on Open Container Initiative (OCI) images via Docker, packages software binaries along with all dependencies into immutable, isolated units. Kubernetes (K8s) serves as the industry standard orchestrator, automating container scheduling, horizontal pod autoscaling, health monitoring, and ingress traffic routing across heterogeneous compute clusters.

Serverless Computing and Event-Driven Design
Function-as-a-Service (FaaS) platforms such as AWS Lambda or Cloudflare Workers, code executes ephemeral in response to discrete triggers, scaling dynamically from zero to thousands of parallel instances within

milliseconds. This model minimizes idle compute costs and fosters event-driven architectures where disparate microservices react asynchronously to business state transitions. Operating reliable distributed systems necessitates continuous observability based on the three pillars: metrics, logs, and distributed traces (via OpenTelemetry). Site Reliability Engineering (SRE) practices leverage multi-region failover, chaos engineering, and circuit breaker patterns to achieve four-nines (99.99%) availability in the presence of inevitable network partitions and hardware failures.